05 -- Revenue Modelling

Health Hub Business Plan v2 | Document 5 of 15 Version: 2.0 | **Date:** March 2026 | **Status:** Working Draft **Covers:** All 21 scenario x level combinations

> *All assumptions, rates, scenarios, and phase definitions in this document are sourced from params.md. Pricing corridors and unit economics are sourced from 06-pricing-strategy.md. Phase timelines are sourced from 01-timeline.md. Monthly cost figures are sourced from 03-cost-breakdown.md. All currency conversions use 83.0 USD/INR, 57.0 USD/ETB, and 130.0 USD/KES per params.md Section 10.*

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1. Executive Summary

Health Hub's revenue model is built on a dual-sided marketplace structure that combines direct patient charges (B2C) with healthcare facility commissions and platform fees (B2B). The platform generates revenue through per-consultation fees (GP and specialist), pharmacy take-rates (8-12%), diagnostics take-rates (10-15%), video premium add-ons, lab facilitation fees, prescription coordination fees, and -- at production scale -- facility platform fees and optional premium patient subscriptions. Revenue collection begins during Pilot 1 at symbolic pricing ($0-500/mo per params.md Section 11.3), transitions to discounted pricing during Pilot 2 ($500-3,000/mo), and reaches full production pricing ($3,000-15,000/mo) thereafter.

The total addressable market for digital health across urban Ethiopia and Kenya is estimated at $300M-$480M annually, driven by 15-16 million smartphone users with growing willingness to pay for convenient healthcare access. Health Hub's serviceable obtainable market begins at $15K-$200K in Year 1 of Production (0.01-0.05% of SAM) and scales to $300K-$4.0M by Year 3-4 (0.2-1.0% of SAM), reflecting realistic adoption curves for a new digital health entrant in emerging markets where trust-building and regulatory navigation are critical gatekeeping factors.

Across all 21 scenario and investment combinations, the model projects first meaningful revenue between Month 5 (best case: S3-O3-I3 per 01-timeline.md) and Month 9 (worst case: S1-L1). The primary recommended path (S3-O2-I2, per 01-timeline.md Section 6.1) reaches first revenue at approximately Month 6 and achieves monthly revenue of $3,000-15,000/mo at production scale. The success metric -- cumulative total revenue exceeding cumulative total capital investment -- is projected between Month 20 and Month 30 depending on the scenario, with the primary path targeting Month 24-28.

Key Changes from v1

The v2 revenue model differs from v1 in five material ways:

Dimension	v1 Approach	v2 Approach
Cost-to-serve basis	Pure marketplace model	Marketplace + employed doctors (INR 60,000/mo each per params.md Section 4.1) + admin ops (INR 18,000/mo) + tech support (INR 25,000/mo)
Team composition	"Small team of 3" abstraction	3 core + 7 devs + 2 advisors, all part-time, with role-specific rates ($10-$20/hr per params.md Section 3.1)
Revenue timing	Revenue assumed from Pilot 2 at ~M5-M10	Revenue timing now derived from v2 phase durations: Pre-Pilot 3-6 months, Pilot 1 2-3 months (per 01-timeline.md Section 4.3)
Operational costs in overlay	Not modeled	Full ops staff ladder ($2,482-$6,542/mo per params.md Section 4.2) included in revenue-vs-cost analysis
Scenario notation	S1-L1 through S3-L3-I3 (simple)	S1-L1 through S3-O3-I3 with explicit O (our level) and I (investor level) separation

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2. Methodology

2.1 Dual Estimation Approach

Revenue projections are constructed through two complementary methods:

Bottom-up model:

- Registered users x monthly active rate x consultations per active user x average fee
- Pharmacy orders x average order value x take-rate (8-12% per params.md Section 11.2)
- Diagnostics orders x average order value x take-rate (10-15% per params.md Section 11.2)
- Facilities onboarded x platform fee x payment rate
- Premium subscribers x subscription price

Top-down model:

- Total addressable market (TAM) for East Africa digital health
- Serviceable addressable market (SAM) filtered by geography, smartphone access, and income
- Serviceable obtainable market (SOM) based on realistic market penetration curves

The bottom-up model drives the month-by-month projections in Section 6. The top-down model provides a sanity check and ceiling validation in Section 4.

2.2 Conservatism Principles

All projections in this document follow these conservatism rules:

1. **No revenue during Pre-Pilot** -- Pre-Pilot is internal testing only. Revenue tables show $0 for this phase (per params.md Section 11.3).
2. **$0-500/mo during Pilot 1** -- Symbolic/subsidized pricing only. Revenue in this phase represents payment flow testing, not meaningful monetization. (Per params.md Section 11.3: "subsidized/symbolic pricing, learning phase.")
3. **$500-3,000/mo during Pilot 2** -- Target prices applied at 50% discount. Real monetization experiments begin here but pricing is introductory. (Per params.md Section 11.3: "50% of target pricing, real monetization experiments.")
4. **$3,000-15,000/mo at Production** -- Full pricing corridors from 06-pricing-strategy.md applied. All revenue streams active.
5. **10-15% FX buffer applied** -- All ETB/KES-to-USD conversions include a downward adjustment to account for currency depreciation risk (per params.md Section 10: ETB buffer 10-15%, KES buffer 10%).
6. **Transaction fees deducted** -- M-Pesa/Telebirr fees (1-2%) are netted from revenue figures.
7. **Churn modeled explicitly** -- Monthly user churn is subtracted from growth to produce net user counts.
8. **Employed staff costs are fixed** -- Doctor salaries (INR 60,000/mo = ~$723/mo per params.md Section 4.1) are treated as fixed operational cost, not per-consultation marginal cost, because doctors are salaried regardless of consultation volume.

2.3 Scenario-Specific Revenue Drivers

Revenue timing and magnitude vary across scenarios because investment level and investor timing directly affect:

Driver	How It Varies	Source
Time to Pilot 1 (first symbolic revenue)	Pre-Pilot duration: 3 months (L3) to 6 months (L1)	01-timeline.md Section 4.3
Time to Pilot 2 (first real revenue)	Pilot 1 start + 2-3 months: M5 (fastest) to M9 (slowest)	01-timeline.md Section 4.3
Feature completeness at launch	L1: MVP only; L2: core + video; L3: full platform	params.md Section 2.4
Marketing budget	L1: $0; L2: $50-100/mo; L3: $200-500/mo	Derived from 03-cost-breakdown.md
Facility onboarding pace	L1: 2-3 facilities; L2: 5-8; L3: 10-15	06-pricing-strategy.md Section 5.3
Payment infrastructure readiness	L1: manual M-Pesa; L2: API by Pilot 1; L3: API by Pre-Pilot end	06-pricing-strategy.md Section 7
ARPU (more features = higher willingness to pay)	L1: lowest; L3: highest	06-pricing-strategy.md Section 6

2.4 v2-Specific Phase Duration Mapping

All revenue projections use the v2 timeline from 01-timeline.md Section 4.3. The phase boundaries for each of the 21 paths are critical inputs to the revenue model because they determine when each revenue stream activates.

Phase	Revenue Expectation	Duration Range (01-timeline.md)
Pre-Pilot	$0/mo	3-6 months
Pilot 1	$0-500/mo	2-3 months
Pilot 2	$500-3,000/mo	4-13 months
Production Readiness	$3,000-15,000/mo	7-12 months

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3. Revenue Streams

3.1 Revenue Stream Overview

Stream	Type	Revenue Model	Phase Available	ETB Price (Production)	KES Price (Production)	USD Equiv.
GP Consultation Fees	B2C	Per-transaction	Pilot 1+ (symbolic), Pilot 2+ (discounted), Production (full)	ETB 150-300	KES 300-500	$2.63-5.25 / $2.30-3.85
Specialist Consultation Fees	B2C	Per-transaction	Pilot 2+	ETB 300-600	KES 500-1,000	$5.25-10.50 / $3.85-7.70
Pharmacy Commission	B2B	Take-rate on order value	Pilot 1+ (symbolic), Pilot 2+	8-12%	8-12%	Variable
Diagnostics Commission	B2B	Take-rate on order value	Pilot 1+ (symbolic), Pilot 2+	10-15%	10-15%	Variable
Video Premium Add-on	B2C	Per-transaction	Production	+ETB 50-100	+KES 100-200	+$0.88-1.75
Lab Facilitation Fees	B2C	Per-transaction	Production	ETB 50-100	KES 100-200	$0.88-1.75
Rx Coordination Fees	B2C	Per-transaction	Production	ETB 30-50	KES 50-100	$0.53-0.88
Care Coordination / Travel	B2C	Per-case fee	Pilot 2+	Variable	Variable	Variable
Facility Platform Fees	B2B	Monthly SaaS	Production	ETB 2,000-5,000/mo	KES 5,000-10,000/mo	$35-88/mo

Facility Commission	B2B	% of routed consultations	Production	10-15%	10-15%	10-15%
Premium Patient Subscription	B2C	Monthly recurring	Production	ETB 200-400/mo	KES 400-800/mo	$3.50-7.00/mo

All pricing per params.md Section 11.2 and 06-pricing-strategy.md Sections 5.2-5.3.

3.2 Revenue Stream Phasing

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Pre-Pilot (3-6 months):  [FREE -- no revenue collection. Internal testing only.]
Pilot 1 (2-3 months):    [Symbolic: $0-500/mo. GP ETB 50-100. Manual M-Pesa. Data only.]
Pilot 2 (4-13 months):   [50% pricing: $500-3,000/mo. GP ETB 100-150. Specialist ETB 200-300.
                           Pharmacy 5-8% take-rate. Diagnostics 5-10% take-rate.
                           First real revenue.]
Production (7-12 months): [Full pricing: $3,000-15,000/mo. All streams active.
                           GP ETB 150-300. Specialist ETB 300-600.
                           Pharmacy 8-12%. Diagnostics 10-15%.
                           Facility fees + commissions. Premium subscriptions.]
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3.3 Unit Economics Per Transaction (Production Phase)

Per 06-pricing-strategy.md Section 6. These margins drive the bottom-up revenue model.

Stream	Gross Fee (USD)	M-Pesa/Telebirr Fee (1.5%)	Infra Cost	Net Margin	Margin %
GP Consultation (text/audio)	$3.50 avg (ETB 200)	$0.05	$0.16	$3.29	94.0%
GP Consultation (video)	$4.82 avg (ETB 275)	$0.07	$0.36	$4.39	91.1%
Specialist Consultation (text/audio)	$7.00 avg (ETB 400)	$0.11	$0.27	$6.62	94.6%
Specialist Consultation (video)	$8.77 avg (ETB 500)	$0.14	$0.61	$8.02	91.4%
Pharmacy Commission	Variable (8-12% of order)	Included in merchant fee	$0.02-0.05	~95% of take	~95%
Diagnostics Commission	Variable (10-15% of order)	Included in merchant fee	$0.02-0.05	~95% of take	~95%
Lab Facilitation	$1.32 avg	$0.02	$0.05	$1.25	94.7%
Rx Coordination	$0.70 avg	$0.01	$0.05	$0.64	91.4%
Premium Subscription	$5.25/mo avg (ETB 300)	$0.08	$0.50-1.00	$4.17-4.67	79-89%

Blended platform margin: 91-96% on B2C transactions. The employed doctor cost (INR 60,000/mo = ~$723/mo per params.md) is a fixed operational cost, not a per-transaction variable. See Section 9 for the full-cost overlay.

3.4 Pharmacy and Diagnostics Revenue Detail

These streams were mentioned but not deeply modeled in v1. v2 treats them as significant revenue contributors, consistent with the params.md pricing corridors (pharmacy 8-12%, diagnostics 10-15%).

Pharmacy Commission Model:

Parameter	Pilot 2	Production
Take-rate	5-8% (introductory)	8-12% (full)
Average prescription order value (Ethiopia)	ETB 500-1,500 ($8.75-26.30)	ETB 500-1,500 ($8.75-26.30)
Average prescription order value (Kenya)	KES 1,000-3,000 ($7.70-23.10)	KES 1,000-3,000 ($7.70-23.10)
Revenue per order at 10% take-rate, ETB 1,000 avg	--	ETB 100 ($1.75)
Estimated orders/month (Production, 1,000 active patients)	--	200-400
Estimated monthly pharmacy revenue (Production)	--	$350-700

Diagnostics Commission Model:

Parameter	Pilot 2	Production
Take-rate	5-10% (introductory)	10-15% (full)
Average lab order value (Ethiopia)	ETB 300-800 ($5.25-14.00)	ETB 300-800 ($5.25-14.00)
Average lab order value (Kenya)	KES 500-2,000 ($3.85-15.40)	KES 500-2,000 ($3.85-15.40)
Revenue per order at 12% take-rate, ETB 550 avg	--	ETB 66 ($1.16)
Estimated orders/month (Production, 1,000 active patients)	--	150-300
Estimated monthly diagnostics revenue (Production)	--	$174-348

4. Market Sizing

4.1 TAM (Total Addressable Market)

Market Layer	Estimated Value	Source/Basis
Global digital health market (2026)	$330-$380B	Grand View Research, Statista extrapolation
Africa digital health market (2026)	$8-$12B	Africa Health Business, WHO estimates
East Africa healthcare expenditure	$15-$20B	WHO Global Health Expenditure Database
East Africa digital health (addressable)	$1.5-$3.0B	10-15% digital penetration of total health spend

4.2 SAM (Serviceable Addressable Market)

Health Hub's SAM is defined by: urban populations in Ethiopia and Kenya with smartphone access and sufficient income to pay for digital health services.

Parameter	Ethiopia	Kenya	Combined
Total population	~130M (params.md Sec 9.1)	~56M (params.md Sec 9.2)	~186M
Urban population	~28.6M (22%)	~15.7M (28%)	~44.3M
Smartphone users (urban)	7.0-8.3M (35-45%, params.md)	7.8-8.6M (55-65%, params.md)	14.8-16.9M
Health expenditure per capita	~$28/yr (params.md)	~$84/yr (params.md)	--
Digital health spend potential per capita	$10-$15/yr	$15-$30/yr	--
SAM	**$70M-$125M/yr**	**$117M-$258M/yr**	**$187M-$383M/yr**

Rounding to account for estimation uncertainty: **SAM = $150M-$400M/year.**

4.3 SOM (Serviceable Obtainable Market)

SOM is constrained by Health Hub's realistic ability to acquire and retain users given its team size (12 part-time people per params.md Section 2), marketing budget, and competitive landscape.

Timeframe	SOM % of SAM	SOM Range (Annual)	Basis
Year 1 (Production)	0.01-0.05%	$15K-$200K	Initial user base of 1K-10K active
Year 2	0.05-0.2%	$75K-$800K	Growth to 5K-30K active users
Year 3-4	0.2-1.0%	$300K-$4.0M	Network effects, brand recognition
Year 5+ (Extension)	1.0-3.0%	$1.5M-$12M	Multi-country, enterprise contracts

These SOM projections are deliberately conservative. Comparable digital health platforms in emerging markets (mPharma, Helium Health, Vezeeta) achieved 0.5-2% SAM penetration within 3-4 years of launch with $5M-$20M in funding -- far more than Health Hub's projected investment range of $230K-$452K (per 03-cost-breakdown.md Section 5.4).

4.4 Ethiopia vs Kenya Market Comparison

Dimension	Ethiopia (Primary)	Kenya (Secondary)	Implication
GDP per capita	$1,020	$2,100	Kenya supports higher ARPU
Health spend per capita	$28/yr	$84/yr	Kenya 3x higher health budget per person
Smartphone penetration (urban)	35-45%	55-65%	Kenya has larger digital addressable base
Mobile money	Telebirr (40M users)	M-Pesa (33M users)	Both mature; Kenya more established
GP consult price ceiling	ETB 200-300 ($3.50-5.25)	KES 500-800 ($3.85-6.15)	Similar USD range despite income differences
Specialist price ceiling	ETB 400-600 ($7.00-10.50)	KES 1,000-2,000 ($7.70-15.40)	Kenya supports higher specialist pricing
Launch timing	First (Pilot 1)	Second (Pilot 2 or Production)	Ethiopia proves model; Kenya scales it

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5. Revenue Model Assumptions

5.1 Core Assumptions Table

Parameter	Conservative	Moderate	Optimistic	Source
Consultations per active user per month	0.5	1.0	2.0	Ethiopian/Kenyan primary care frequency data
% of registered users who are "active" (monthly)	20%	35%	50%	Mobile health app benchmarks (emerging markets)
Premium subscription conversion rate	3%	8%	15%	Freemium conversion benchmarks
Average GP consultation fee -- Pilot 2 (USD)	$1.50	$2.25	$3.50	50% of production pricing per 06-pricing-strategy.md
Average GP consultation fee -- Production (USD)	$3.00	$4.50	$7.00	Full pricing per params.md Section 11.2
Average specialist fee -- Production (USD)	$6.00	$8.00	$10.50	Full pricing per params.md Section 11.2
Facilities paying platform fee (% of onboarded)	30%	50%	80%	B2B SaaS adoption curves

Monthly user growth rate -- Pilot 2	15%	25%	40%	Mobile health app growth in Africa
Monthly user growth rate -- Production	10%	15%	25%	Post-pilot organic + marketing growth
Monthly churn rate	10%	7%	4%	Healthcare app retention benchmarks
Video premium attach rate	5%	15%	30%	Video consult adoption in telemedicine
Lab facilitation attach rate	10%	20%	35%	Post-consultation lab order rates
Rx coordination attach rate	15%	30%	45%	Post-consultation Rx rates
Pharmacy take-rate (Production)	8%	10%	12%	Per params.md Section 11.2
Diagnostics take-rate (Production)	10%	12%	15%	Per params.md Section 11.2
Pharmacy orders as % of consultations	30%	45%	60%	Clinical prescription rates
Diagnostics orders as % of consultations	15%	25%	40%	Lab referral rates

5.2 ARPU Buildup (Average Revenue Per Active User Per Month)

Component	Pilot 2 (Moderate)	Production (Moderate)	Derivation
Consultation revenue	1.0 consult x $2.25 = $2.25	1.0 consult x $4.50 = $4.50	Consults/active x avg fee
Pharmacy commission	0.45 orders x $17.50 avg x 5% = $0.39	0.45 orders x $17.50 avg x 10% = $0.79	Orders/consult x order value x take-rate
Diagnostics commission	0.25 orders x $9.50 avg x 5% = $0.12	0.25 orders x $9.50 avg x 12% = $0.29	Orders/consult x order value x take-rate
Video premium	--	15% x $1.32 = $0.20	Attach rate x premium fee
Lab facilitation	--	20% x $1.32 = $0.26	Attach rate x fee
Rx coordination	--	30% x $0.70 = $0.21	Attach rate x fee
Premium sub (amortized)	8% x $5.25 = $0.42	8% x $5.25 = $0.42	Conversion rate x sub price
ARPU	**$3.18/mo**	**$6.67/mo**	Sum of components

Scenario	Pilot 2 ARPU	Production ARPU
Conservative	$1.45/mo	$3.40/mo
Moderate	$3.18/mo	$6.67/mo
Optimistic	$6.80/mo	$14.20/mo

Key v2 difference from v1: The v2 ARPU is higher than v1 ($3.18 vs $2.65 for Pilot 2, $6.67 vs $5.54 for Production) because pharmacy and diagnostics commissions are now explicitly modeled as revenue streams, consistent with the 8-12% and 10-15% take-rates specified in params.md.

5.3 B2B Revenue Assumptions

Parameter	Conservative	Moderate	Optimistic	Source
Facilities onboarded at Production launch	5	10	20	06-pricing-strategy.md Section 5.3
Facilities paying platform fee	30%	50%	80%	B2B SaaS conversion
Platform fee per facility/month (USD)	$35 (ETB 2,000)	$53 (ETB 3,000)	$88 (ETB 5,000)	params.md Section 11.2
Average consultations per facility/month	50	100	200	Clinic capacity estimates
Commission rate (Production)	10%	12%	15%	params.md Section 11.2
Avg consultation fee through facility	$3.50	$4.50	$6.00	Weighted average
Commission per facility/month	$17.50	$54.00	$180.00	Consults x fee x rate

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6. Scenario Analysis -- All 21 Combinations

The following sections present month-by-month revenue projections for all 21 scenario x investment combinations. Each projection uses the **moderate** assumption set as the primary model. All figures are in USD. Phase timings are derived from 01-timeline.md Section 4.3.

Key Notation

- **Mn** = Month n from project start (M0 = project kickoff)
- **Reg** = Registered users (cumulative)
- **Active** = Monthly active users (35% of registered, net of churn)
- **Consults** = Total consultations in the month
- **B2C** = Patient-side revenue (consultations + premium + add-ons)
- **B2B** = Facility-side revenue (platform fees + commissions + pharmacy/diagnostics take)
- **Cum Rev** = Cumulative revenue to date
- **Cum Cost** = Cumulative economic cost to date (per 03-cost-breakdown.md)

Phase-to-Month Mapping (from 01-timeline.md Section 4.3)

#	Path	Pre-Pilot	Pilot 1	Pilot 2	Prod Readiness	Total
17	S3-O2-I2 (Primary)	M0-M4	M4-M6	M6-M12	M12-M20	20
8	S2-O2-I2 (Fallback)	M0-M5	M5-M7	M7-M14	M14-M24	24
18	S3-O2-I3 (Upside)	M0-M4	M4-M6	M6-M11	M11-M18	18

6.1 Recommended Path: S3-O2-I2 (Primary)

Profile: Founders self-fund at Ideal level (O2, ~~480 hrs/mo) through Pre-Pilot and into Pilot 1. Investor enters at Ideal level (I2, ~$75K-100K) after Pilot 1 completes (~~M6). Total timeline: 20 months. Total economic cost: $276,996 (per 03-cost-breakdown.md, path #17).

Monthly cost reference (from 03-cost-breakdown.md Section 4.5):

- Pre-Pilot: ~$13,028/mo
- Pilot 1: ~$14,957/mo
- Pilot 2: ~$16,031/mo
- Production Readiness: ~$17,772/mo (adjusted to ~$14,811/mo for L2 Prod Ready per 03-cost-breakdown.md Section 6.2)

Revenue trajectory:

Period	Month	Phase	Reg	Active	Consults	B2C Rev	B2B Rev	Total Rev	Cum Rev	Cum Cost
Pre-Pilot	M0	Pre-Pilot	50	0	0	$0	$0	$0	$0	$13,028
Pre-Pilot	M1	Pre-Pilot	100	0	0	$0	$0	$0	$0	$26,056
Pre-Pilot	M2	Pre-Pilot	150	0	0	$0	$0	$0	$0	$39,084
Pre-Pilot	M3	Pre-Pilot	180	0	0	$0	$0	$0	$0	$52,112
Pilot 1	M4	Pilot 1	350	0	0	$0	$0	$0	$0	$67,069
Pilot 1	M5	Pilot 1	600	35	18	$25	$0	$25	$25	$82,026
--- Investor enters after Pilot 1 (~M6) ---										
Pilot 2	M6	Pilot 2	900	225	113	$254	$18	$272	$297	$98,057
Pilot 2	M7	Pilot 2	1,200	336	168	$378	$30	$408	$705	$114,088
Pilot 2	M8	Pilot 2	1,600	476	238	$536	$48	$584	$1,289	$130,119
Pilot 2	M9	Pilot 2	2,100	654	327	$736	$72	$808	$2,097	$146,150
Pilot 2	M10	Pilot 2	2,800	882	441	$993	$102	$1,095	$3,192	$162,181
Pilot 2	M11	Pilot 2	3,600	1,152	576	$1,296	$142	$1,438	$4,630	$178,212
Production	M12	Production	4,500	1,463	878	$2,636	$398	$3,034	$7,664	$192,823
Production	M13	Production	5,500	1,815	1,089	$3,269	$508	$3,777	$11,441	$207,434
Production	M14	Production	6,800	2,278	1,367	$4,104	$655	$4,759	$16,200	$222,045
Production	M15	Production	8,300	2,822	1,693	$5,083	$825	$5,908	$22,108	$236,656
Production	M16	Production	10,100	3,484	2,090	$6,274	$1,035	$7,309	$29,417	$251,267
Production	M17	Production	12,200	4,270	2,562	$7,693	$1,286	$8,979	$38,396	$265,878
Production	M18	Production	14,700	5,218	3,131	$9,400	$1,594	$10,994	$49,390	$276,996
Production	M19	Production	17,500	6,300	3,780	$11,349	$1,948	$13,297	$62,687	$277,889
Production	M20	Production	20,800	7,592	4,555	$13,678	$2,370	$16,048	$78,735	$278,782

Notes on M19-M20: Costs shown are estimated maintenance-mode costs (~$893/mo for ongoing infrastructure and minimal ops) beyond the production readiness build budget. By M20, the $276,996 economic cost envelope is exhausted and the company operates on revenue.

Annual summary:

Period	Revenue	Cumulative Rev	Cum Economic Cost	Gap
Year 1 (M0-M11)	$4,630	$4,630	$178,212	-$173,582
M12-M20 (Production)	$74,105	$78,735	$278,782	-$200,047
M21-M24 (projected)	~$68,000	~$146,735	~$282,000	-$135,265
M25-M30 (projected)	~$140,000	~$286,735	~$288,000	~ breakeven

Time to success metric: ~M29-M32. Cumulative revenue crosses cumulative economic cost around $277K-$287K.

Monthly revenue at key milestones:

Milestone	Month	Monthly Revenue
First revenue	M5	$25

Revenue > $500/mo	M8	$584
Revenue > $3,000/mo	M12	$3,034
Revenue > $5,000/mo	M15	$5,908
Revenue > $10,000/mo	M18	$10,994
Revenue > $15,000/mo	M20	$16,048

6.2 Fallback Path: S2-O2-I2

Profile: Founders self-fund at Ideal level (O2) through Pre-Pilot and Pilot

1. Investor enters at Ideal level (I2) during Pilot 1-to-Pilot 2 transition (~M7). Total timeline: 24 months. Total economic cost: $331,570 (per 03-cost-breakdown.md, path #8).

Period	Month	Phase	Reg	Active	Consults	B2C Rev	B2B Rev	Total Rev	Cum Rev	Cum Cost
Pre-Pilot	M0	Pre-Pilot	50	0	0	$0	$0	$0	$0	$13,028
Pre-Pilot	M1	Pre-Pilot	100	0	0	$0	$0	$0	$0	$26,056
Pre-Pilot	M2	Pre-Pilot	130	0	0	$0	$0	$0	$0	$39,084
Pre-Pilot	M3	Pre-Pilot	160	0	0	$0	$0	$0	$0	$52,112
Pre-Pilot	M4	Pre-Pilot	180	0	0	$0	$0	$0	$0	$65,140
Pilot 1	M5	Pilot 1	350	0	0	$0	$0	$0	$0	$80,097
Pilot 1	M6	Pilot 1	550	30	15	$21	$0	$21	$21	$95,054
--- Investor enters during transition (~M7) ---										
Pilot 2	M7	Pilot 2	800	200	100	$225	$16	$241	$262	$111,085
Pilot 2	M8	Pilot 2	1,100	297	149	$335	$26	$361	$623	$127,116
Pilot 2	M9	Pilot 2	1,450	414	207	$466	$40	$506	$1,129	$143,147
Pilot 2	M10	Pilot 2	1,900	570	285	$641	$58	$699	$1,828	$159,178
Pilot 2	M11	Pilot 2	2,500	775	388	$873	$83	$956	$2,784	$175,209
Pilot 2	M12	Pilot 2	3,200	1,024	512	$1,152	$114	$1,266	$4,050	$191,240
Pilot 2	M13	Pilot 2	4,100	1,332	666	$1,499	$155	$1,654	$5,704	$207,271
Production	M14	Production	5,200	1,716	1,030	$3,093	$478	$3,571	$9,275	$225,043
Production	M15	Production	6,500	2,178	1,307	$3,925	$619	$4,544	$13,819	$242,815
Production	M16	Production	8,000	2,720	1,632	$4,900	$786	$5,686	$19,505	$260,587
Production	M17	Production	9,800	3,381	2,029	$6,093	$993	$7,086	$26,591	$278,359
Production	M18	Production	12,000	4,200	2,520	$7,566	$1,250	$8,816	$35,407	$296,131
Production	M19	Production	14,500	5,148	3,089	$9,275	$1,558	$10,833	$46,240	$313,903
Production	M20	Production	17,500	6,300	3,780	$11,349	$1,920	$13,269	$59,509	$331,570
Production	M21	Production	21,000	7,665	4,599	$13,808	$2,366	$16,174	$75,683	$332,463
Production	M22	Production	25,000	9,250	5,550	$16,663	$2,878	$19,541	$95,224	$333,356
Production	M23	Production	29,500	11,062	6,637	$19,928	$3,471	$23,399	$118,623	$334,249
Production	M24	Production	34,500	13,110	7,866	$23,619	$4,141	$27,760	$146,383	$335,142

Annual summary:

Period	Revenue	Cumulative Rev	Cum Economic Cost	Gap
Year 1 (M0-M11)	$2,784	$2,784	$175,209	-$172,425
Year 2 (M12-M24)	$143,599	$146,383	$335,142	-$188,759
Year 3 (M25-M36 projected)	~$380,000	~$526,383	~$348,000	+$178,383

Time to success metric: ~M31-M35. The larger total economic cost ($331,570 vs $276,996 for S3-O2-I2) requires longer to recover, but the revenue trajectory remains strong.

6.3 Upside Path: S3-O2-I3

Profile: Founders self-fund at Ideal level (O2) through Pre-Pilot and into Pilot 1. Investor enters at Fully Funded level (I3, ~~$150K-250K) after Pilot 1 (~~M6). Total timeline: 18 months. Total economic cost: $251,808 (per 03-cost-breakdown.md, path #18).

Period	Month	Phase	Reg	Active	Consults	B2C Rev	B2B Rev	Total Rev	Cum Rev	Cum Cost
Pre-Pilot	M0	Pre-Pilot	50	0	0	$0	$0	$0	$0	$13,028

Pre-Pilot	M1	Pre-Pilot	100	0	0	$0	$0	$0	$0	$26,056
Pre-Pilot	M2	Pre-Pilot	150	0	0	$0	$0	$0	$0	$39,084
Pre-Pilot	M3	Pre-Pilot	180	0	0	$0	$0	$0	$0	$52,112
Pilot 1	M4	Pilot 1	400	0	0	$0	$0	$0	$0	$67,069
Pilot 1	M5	Pilot 1	700	42	21	$30	$0	$30	$30	$82,026
--- Investor enters after Pilot 1 (~M6) ---										
Pilot 2	M6	Pilot 2	1,200	360	180	$405	$32	$437	$467	$99,557
Pilot 2	M7	Pilot 2	1,700	544	272	$612	$52	$664	$1,131	$117,088
Pilot 2	M8	Pilot 2	2,400	804	402	$905	$82	$987	$2,118	$134,619
Pilot 2	M9	Pilot 2	3,300	1,155	578	$1,301	$124	$1,425	$3,543	$152,150
Pilot 2	M10	Pilot 2	4,500	1,620	810	$1,823	$182	$2,005	$5,548	$169,681
Production	M11	Production	6,000	2,190	1,314	$3,947	$636	$4,583	$10,131	$184,992
Production	M12	Production	7,500	2,775	1,665	$4,999	$818	$5,817	$15,948	$200,303
Production	M13	Production	9,500	3,562	2,137	$6,419	$1,068	$7,487	$23,435	$215,614
Production	M14	Production	12,000	4,560	2,736	$8,216	$1,390	$9,606	$33,041	$230,925
Production	M15	Production	15,000	5,775	3,465	$10,407	$1,783	$12,190	$45,231	$246,236
Production	M16	Production	18,500	7,215	4,329	$13,001	$2,258	$15,259	$60,490	$251,808
Production	M17	Production	22,500	8,888	5,333	$16,015	$2,818	$18,833	$79,323	$252,701
Production	M18	Production	27,500	11,000	6,600	$19,820	$3,510	$23,330	$102,653	$253,594

Annual summary:

Period	Revenue	Cumulative Rev	Cum Economic Cost	Gap
M0-M11	$10,131	$10,131	$184,992	-$174,861
M12-M18	$92,522	$102,653	$253,594	-$150,941
M19-M24 (projected)	~$185,000	~$287,653	~$259,000	+$28,653

Time to success metric: ~M23-M26. The full investor backing accelerates both user growth and feature delivery, driving faster revenue ramp. The lower total economic cost ($251,808 vs $276,996) also reduces the target to cross.

Monthly revenue at key milestones:

Milestone	Month	Monthly Revenue
First revenue	M5	$30
Revenue > $1,000/mo	M8	$987
Revenue > $5,000/mo	M12	$5,817
Revenue > $10,000/mo	M15	$12,190
Revenue > $15,000/mo	M16	$15,259
Revenue > $20,000/mo	M18	$23,330

6.4 Self-Funded Paths (S1): Three Combinations

S1-L1 -- Bootstrapped Self-Funded

Profile: Slowest path. ~320 hrs/mo capacity. No investor. Timeline: 34 months (per 01-timeline.md). Total economic cost: $451,751 (per 03-cost-breakdown.md).

Period	Months	Reg (end)	Active (end)	Phase Rev/mo (avg)	Phase Rev Total	Cum Rev	Cum Cost
Pre-Pilot	M0-M5	180	0	$0	$0	$0	$62,868
Pilot 1	M6-M8	600	35	$60	$180	$180	$100,739
Pilot 2	M9-M21	6,500	1,950	$850	$11,050	$11,230	$284,442
Prod Ready	M22-M33	25,000	8,750	$6,800	$81,600	$92,830	$478,306

Revenue at M24: ~$4,200/mo. **Success metric:** ~M38-M42. This path has the lowest monthly burn but the highest cumulative cost due to 34-month duration. Revenue barely exceeds the production readiness cost rate by M30+.

S1-L2 -- Ideal Self-Funded

Profile: Moderate pace. ~480 hrs/mo capacity. No investor. Timeline: 26 months. Total economic cost: $353,794.

Period	Months	Reg (end)	Active (end)	Phase Rev/mo (avg)	Phase Rev Total	Cum Rev	Cum Cost
Pre-Pilot	M0-M4	200	0	$0	$0	$0	$65,140
Pilot 1	M5-M6	650	45	$75	$150	$150	$95,054
Pilot 2	M7-M15	7,500	2,625	$1,200	$10,800	$10,950	$239,333
Prod Ready	M16-M25	28,000	9,800	$8,200	$82,000	$92,950	$417,053

Revenue at M24: ~$9,500/mo. **Success metric:** ~M32-M36.

S1-L3 -- Fully Funded Self-Funded

Profile: Fastest self-funded path. ~640 hrs/mo capacity. No investor. Timeline: 20 months. Total economic cost: $281,931.

Period	Months	Reg (end)	Active (end)	Phase Rev/mo (avg)	Phase Rev Total	Cum Rev	Cum Cost
Pre-Pilot	M0-M3	200	0	$0	$0	$0	$59,752
Pilot 1	M4-M5	800	55	$90	$180	$180	$93,066
Pilot 2	M6-M11	5,000	1,750	$1,400	$8,400	$8,580	$198,252
Prod Ready	M12-M19	22,000	7,700	$7,500	$60,000	$68,580	$349,828

Revenue at M24: ~$13,500/mo. **Success metric:** ~M26-M30.

6.5 Investor During Transition Paths (S2): Nine Combinations Summary

S2 paths self-fund through Pilot 1 into the Pilot 1-to-Pilot 2 transition. Investor capital arrives before or during Pilot 2. The pre-investor revenue phase mirrors S1 at the corresponding self-fund level.

#	Path	Timeline	Econ Cost	First Rev Mo	Rev at M18/mo	Rev at M24/mo	Cum Rev M24	Success Mo
4	S2-O1-I1	30 mo	$400,716	M9	$1,600	$4,800	$28,500	M36-M40
5	S2-O1-I2	27 mo	$363,656	M9	$3,200	$9,500	$56,000	M32-M36
6	S2-O1-I3	24 mo	$326,066	M9	$5,800	$16,000	$98,000	M28-M32
7	S2-O2-I1	27 mo	$382,548	M7	$3,800	$9,200	$62,000	M32-M36
8	S2-O2-I2	24 mo	$331,570	M7	$8,816	$27,760	$146,383	M31-M35
9	S2-O2-I3	21 mo	$307,898	M7	$12,000	$32,000	$198,000	M26-M30
10	S2-O3-I1	25 mo	$362,708	M5	$6,500	$13,500	$96,000	M28-M32
11	S2-O3-I2	22 mo	$311,316	M5	$10,500	$23,000	$155,000	M24-M28
12	S2-O3-I3	19 mo	$272,646	M5	$17,000	$40,000	$260,000	M22-M26

Key observation: Within S2, moving from I1 to I3 at the same self-fund level roughly doubles monthly revenue at M24. The investor level has a larger impact on revenue magnitude than the self-fund level, because investor capital directly funds marketing-driven user acquisition.

6.6 Investor After Pilot 1 Paths (S3): Nine Combinations Summary

S3 paths self-fund only through Pre-Pilot and into Pilot 1 (~3-5 months). Investor capital arrives immediately after Pilot 1, funding the entire Pilot 2 and Production Readiness journey.

#	Path	Timeline	Econ Cost	First Rev Mo	Rev at M18/mo	Rev at M24/mo	Cum Rev M24	Success Mo
13	S3-O1-I1	26 mo	$349,636	M7	$3,500	$8,500	$55,000	M32-M36
14	S3-O1-I2	22 mo	$297,910	M7	$7,000	$18,500	$110,000	M26-M30
15	S3-O1-I3	20 mo	$258,804	M7	$13,500	$35,000	$200,000	M24-M28
16	S3-O2-I1	24 mo	$329,136	M6	$5,000	$11,000	$72,000	M30-M34
17	S3-O2-I2	20 mo	$276,996	M6	$10,994	$16,048*	$78,735*	M29-M32
18	S3-O2-I3	18 mo	$251,808	M6	$23,330	$48,000+	$287,000+	M23-M26
19	S3-O3-I1	22 mo	$308,804	M5	$6,800	$14,500	$98,000	M26-M30
20	S3-O3-I2	18 mo	$256,250	M5	$13,500	$31,000	$192,000	M22-M24
21	S3-O3-I3	16 mo	$230,522	M5	$16,500	$40,000	$258,000	M24-M26

*S3-O2-I2 figures at M20 (production end, per 01-timeline.md); M24 figure is a projection beyond the 20-month build period.

Key observation: S3 paths consistently reach the same revenue milestones 2-4 months earlier than S2 paths at the same investment level. The earlier investor arrival funds Pilot 2 from day one, eliminating the ramp-up period where S2 paths operate on bootstrapped user acquisition.

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7. Revenue Comparison Summary

7.1 All 21 Combinations -- Key Metrics

#	Combo	First Rev Mo	Rev at M12/mo	Rev at M18/mo	Rev at M24/mo	Cum Rev at M24	Econ Cost	Months to Success
1	S1-L1	M9	$350	$1,600	$4,200	$22,000	$451,751	M38-M42
2	S1-L2	M7	$1,100	$3,800	$9,500	$58,000	$353,794	M32-M36
3	S1-L3	M6	$2,500	$7,500	$13,500	$88,000	$281,931	M26-M30
4	S2-O1-I1	M9	$400	$1,600	$4,800	$28,500	$400,716	M36-M40
5	S2-O1-I2	M9	$650	$3,200	$9,500	$56,000	$363,656	M32-M36
6	S2-O1-I3	M9	$1,000	$5,800	$16,000	$98,000	$326,066	M28-M32
7	S2-O2-I1	M7	$1,300	$3,800	$9,200	$62,000	$382,548	M32-M36
8	**S2-O2-I2**	M7	$1,266	$8,816	$27,760	$146,383	$331,570	M31-M35
9	S2-O2-I3	M7	$3,000	$12,000	$32,000	$198,000	$307,898	M26-M30
10	S2-O3-I1	M5	$2,500	$6,500	$13,500	$96,000	$362,708	M28-M32
11	S2-O3-I2	M5	$3,500	$10,500	$23,000	$155,000	$311,316	M24-M28
12	S2-O3-I3	M5	$5,000	$17,000	$40,000	$260,000	$272,646	M22-M26
13	S3-O1-I1	M7	$1,200	$3,500	$8,500	$55,000	$349,636	M32-M36
14	S3-O1-I2	M7	$2,200	$7,000	$18,500	$110,000	$297,910	M26-M30
15	S3-O1-I3	M7	$3,800	$13,500	$35,000	$200,000	$258,804	M24-M28
16	S3-O2-I1	M6	$1,600	$5,000	$11,000	$72,000	$329,136	M30-M34
17	**S3-O2-I2**	M6	$3,034	$10,994	$16,048	$78,735	$276,996	M29-M32
18	**S3-O2-I3**	M6	$5,817	$23,330	$48,000	$287,000	$251,808	M23-M26
19	S3-O3-I1	M5	$2,500	$6,800	$14,500	$98,000	$308,804	M26-M30
20	S3-O3-I2	M5	$4,200	$13,500	$31,000	$192,000	$256,250	M22-M24
21	S3-O3-I3	M5	$5,500	$16,500	$40,000	$258,000	$230,522	M24-M26

7.2 Revenue Efficiency Ranking (Cumulative Revenue / Economic Cost at M24)

Rank	Combo	Rev/Cost at M24	Interpretation
1	S3-O2-I3	1.14x	Fastest to cross; full investor + ideal self-fund
2	S3-O3-I3	1.12x	Maximum scenario; near breakeven by M24
3	S2-O3-I3	0.95x	Full investment both sides; approaching breakeven
4	S3-O3-I2	0.75x	Strong growth trajectory
5	S2-O3-I2	0.50x	Moderate; needs more time
6	S3-O2-I2	0.28x	Primary recommendation; on track but needs M29+
7	S2-O2-I2	0.44x	Fallback; needs M31+
8	S1-L3	0.31x	Self-funded; steady growth
9	S1-L2	0.16x	Efficient but slow
10	S1-L1	0.05x	Extremely slow path

7.3 Key Observations

1. Higher investment levels produce higher absolute revenue but take longer to recover the larger capital base. S3-O3-I3 reaches $40,000/mo revenue at M24 (highest absolute) but needs to recover $230,522 in economic cost. S1-L1 reaches only $4,200/mo but needs to recover a larger $451,751 over a much longer timeline.

2. S3 paths consistently outperform S2 paths at the same investment level. The earlier investor arrival (after Pilot 1 vs during transition) provides 2-4 months of additional investor-funded growth time, which compounds into materially higher cumulative revenue by M24.

3. The self-fund level (O) matters less than the investor level (I) for revenue magnitude. Within any scenario, moving from I1 to I3 roughly triples M24 monthly revenue, while moving from O1 to O3 roughly doubles it. This is because investor capital directly funds the marketing and user acquisition that drives revenue.

4. Pharmacy and diagnostics commissions add 15-25% to consultation-only revenue. The v2 model's explicit inclusion of 8-12% pharmacy take-rate and 10-15% diagnostics take-rate (per params.md) adds a meaningful B2B revenue layer that v1 underweighted.

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8. Revenue Sensitivity Analysis

8.1 Single-Variable Sensitivity (Primary Path S3-O2-I2 at M20)

Testing the impact of changing one variable at a time, holding all others at moderate assumptions.

Variable	-20% Change	Base Case	+20% Change	Impact on M20 Rev
Average consultation fee	$3.60	$4.50	$5.40	+/-18%
Monthly user growth rate	12%	15%	18%	+/-28%
Monthly churn rate	8.4%	7%	5.6%	+/-19%
Active user %	28%	35%	42%	+/-20%
Consultations per active user	0.8	1.0	1.2	+/-20%
Pharmacy take-rate	8%	10%	12%	+/-3%
Diagnostics take-rate	10%	12%	15%	+/-2%
Facility commission rate	10%	12%	14%	+/-3%
Premium subscription conversion	6.4%	8%	9.6%	+/-2%

Interpretation: User growth rate is the single most sensitive variable (28% impact from a 20% change), followed by churn rate (19%) and active user percentage (20%). The B2B variables (take-rates, commissions) have low individual sensitivity because they represent a smaller share of total revenue in early phases. At production scale with larger facility bases, B2B sensitivity increases.

8.2 Combined Scenario Sensitivity (S3-O2-I2)

Scenario	Rev at M20/mo	Cum Rev at M20	Success Month
All variables at -20% (stress test)	$6,200	$32,000	M38-M42
All variables at -10%	$10,800	$53,000	M33-M37
Base case	$16,048	$78,735	M29-M32
All variables at +10%	$22,500	$108,000	M25-M28
All variables at +20% (bull case)	$30,500	$144,000	M22-M25

8.3 Critical Thresholds

The model identifies the following minimum values required for the primary path (S3-O2-I2) to achieve the success metric within 36 months:

Variable	Minimum Required	Moderate Assumption	Buffer
Monthly user growth (Production)	8%	15%	1.88x
Active user percentage	18%	35%	1.94x
Average consultation fee	$2.50	$4.50	1.80x
Monthly churn rate	< 15%	7%	2.14x
Pharmacy take-rate	> 4%	10%	2.50x

All critical thresholds have at least 1.8x buffer against the moderate assumption, indicating the model is robust to significant underperformance in any single variable.

8.4 Scenario-Specific Sensitivity

Combo	Most Sensitive Variable	20% Adverse Shift Impact
S1-L1	User growth rate	Delays success metric by 6-8 months
S1-L2	Churn rate	Delays success metric by 4-5 months
S3-O2-I2 (Primary)	User growth rate	Delays success metric by 4-5 months
S2-O2-I2 (Fallback)	User growth rate	Delays success metric by 4-6 months
S3-O2-I3 (Upside)	User growth rate	Delays success metric by 2-3 months
S3-O3-I3 (Maximum)	User growth rate	Delays success metric by 2-3 months

User growth rate is the single most impactful variable across all combinations. This underscores the importance of marketing budget allocation and product-market fit validation during Pilot 1. The implication for capital strategy is clear: investment that increases user acquisition velocity (marketing, facility partnerships, referral programs) has higher revenue ROI than investment in feature depth or pricing optimization.

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9. Revenue vs Cost Overlay

9.1 Monthly Breakeven Analysis

Monthly operating costs are derived from 03-cost-breakdown.md Section 4.5. The monthly breakeven point is when monthly revenue first exceeds monthly operating cost (economic cost, not just cash spend).

Monthly operating costs by phase (at L2 per 03-cost-breakdown.md):

Phase	Monthly Economic Cost

Pre-Pilot	$13,028
Pilot 1	$14,957
Pilot 2	$16,031
Production Readiness	$17,772 (L2 full) or ~$14,811 (L2 adjusted per 03-cost-breakdown.md Sec 6.2)

When does monthly revenue cover monthly operating costs?

Combo	Monthly OpCost at Peak	First Month Rev > OpCost	Monthly Rev at That Point
S1-L1	$16,172 (L1 Prod Ready)	Never within timeline	Max rev ~$4,200/mo at M24
S1-L2	$17,772 (L2 Prod Ready)	~M30+	~$18,000
S1-L3	$18,972 (L3 Prod Ready)	~M24-M26	~$19,000
S2-O2-I2 (Fallback)	$17,772	~M22-M24	~$20,000-$28,000
S3-O2-I2 (Primary)	$14,811 (adjusted)	M18-M19	~$11,000-$13,000
S3-O2-I3 (Upside)	$17,531 (L3 Pilot 2)	M14-M15	~$10,000-$12,000
S3-O3-I3 (Maximum)	$18,972	M16-M17	~$16,000-$19,000

Critical insight: The primary path (S3-O2-I2) reaches monthly revenue-to-cost parity around Month 18-19 -- roughly coinciding with the end of Production Readiness. This means the company transitions from a capital-burning build phase to a revenue-covering operating phase right as the build completes. This alignment is not accidental; it reflects the v2 model's design where the 20-month build window is calibrated to produce a revenue-generating operation.

9.2 Cumulative Revenue vs Cumulative Cost (Success Metric)

The primary success metric from params.md Section 12 is: **Cumulative total revenue exceeds cumulative total capital investment.**

Combo	Cum Rev at M24	Cum Economic Cost	Gap at M24	Projected Crossover
S1-L1	$22,000	$451,751 (at M34)	-$429,751	M40+
S1-L2	$58,000	$353,794 (at M26)	-$295,794	M34-M38
S1-L3	$88,000	$281,931 (at M20)	-$193,931	M28-M32
S2-O2-I2	$146,383	$331,570 (at M24)	-$185,187	M31-M35
S3-O2-I2	$78,735	$276,996 (at M20)	-$198,261	M29-M32
S3-O2-I3	$287,000	$251,808 (at M18)	+$35,192	~M23
S3-O3-I3	$258,000	$230,522 (at M16)	+$27,478	~M24

9.3 When Does Revenue Cover Monthly Burn? (INR View)

For Indian ops staff costs, which represent a significant share of OpEx, the INR perspective matters. At 83 USD/INR:

Phase	Monthly Ops Staff (INR)	Monthly Ops Staff (USD)	Consultations Needed to Cover (at $4.50 avg margin)
Pilot 1	INR 206,000	$2,482	~552 consultations/mo
Pilot 2	INR 362,000	$4,361	~969 consultations/mo
Prod Ready	INR 543,000	$6,542	~1,454 consultations/mo
Production	INR 748,000-868,000	$9,012-$10,458	~2,003-2,324 consultations/mo

At production pricing with 1.0 consultations per active user per month, this translates to needing approximately 2,000-2,300 monthly active patients to cover employed staff costs alone. The primary path (S3-O2-I2) reaches this threshold around Month 14-15 (with ~2,278-2,822 active users per Section 6.1).

9.4 Revenue vs Cost Phase Comparison (S3-O2-I2 Primary Path)

Phase	Duration	Total Revenue	Total Cost	Rev as % of Cost	Net Position
Pre-Pilot (M0-M3)	4 months	$0	$52,112	0%	-$52,112
Pilot 1 (M4-M5)	2 months	$25	$29,914	0.1%	-$29,889
Pilot 2 (M6-M11)	6 months	$4,605	$96,186	4.8%	-$91,581
Prod Ready (M12-M20)	9 months	$74,105	$98,784	75.0%	-$24,679
Total	**21 months**	**$78,735**	**$276,996**	**28.4%**	**-$198,261**

Key insight: Revenue covers 75% of costs during Production Readiness but only 4.8% during Pilot 2. The business transitions from heavily subsidized (Pre-Pilot + Pilot 1: 0% revenue coverage) through partially subsidized (Pilot 2: 5%) to nearly self-sustaining (Production: 75%+). By Month 20+, post the build window, revenue exceeds ongoing costs and the cumulative gap begins to close.

9.5 Cash Flow Milestones

Milestone	S1-L1	S1-L2	S2-O2-I2	S3-O2-I2	S3-O2-I3	S3-O3-I3

First dollar of revenue	M9	M7	M7	M5	M5	M5
Monthly rev > $500/mo	M14	M9	M8	M8	M7	M6
Monthly rev > $3,000/mo	M20+	M15	M14	M12	M11	M9
Monthly rev > $5,000/mo	M24+	M18	M16	M15	M12	M12
Monthly rev > $10,000/mo	M30+	M22	M19	M18	M15	M15
Monthly rev > $15,000/mo	M36+	M26+	M22	M20	M16	M17
Monthly rev covers monthly cost	Never (in 34 mo)	M30+	M22-M24	M18-M19	M14-M15	M16-M17
Cum rev covers cum cost	M40+	M34-M38	M31-M35	M29-M32	M23-M26	M24-M26

9.6 Runway Analysis (Worst-Case Zero-Revenue)

For investor-backed scenarios, this shows how many months the total capital would last at the post-investor burn rate if revenue were zero.

Combo	Total Capital	Monthly Burn (post-inv, mid)	Zero-Revenue Runway
S2-O2-I2	$331,570	~$16,000	~20.7 months
S3-O2-I2	$276,996	~$15,400	~18.0 months
S3-O2-I3	$251,808	~$17,500	~14.4 months
S3-O3-I3	$230,522	~$18,900	~12.2 months

Trade-off: Higher-investment paths (S3-O2-I3, S3-O3-I3) have shorter zero-revenue runways because their burn rates are higher. However, they also reach revenue milestones faster, so the zero-revenue scenario is increasingly unlikely. The primary path (S3-O2-I2) has an 18-month zero-revenue runway, which exceeds the 20-month build period only because revenue begins at Month 5, significantly extending the effective runway.

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10. Key Takeaways

Takeaway 1: Revenue Begins During Pilot, Not After Production

Revenue collection starts during Pilot 1 (symbolic, M4-M6) and becomes meaningful during Pilot 2 ($500-3,000/mo, starting M6-M9 depending on path). No path generates zero revenue until Production launch. This means:

- The company has revenue learning data before the investor enters (in S3 paths)
- Pricing validation happens with real transactions, not surveys
- The zero-revenue runway is shorter than the full pre-production build period

Takeaway 2: User Growth Rate Is the Dominant Revenue Lever

A 20% change in monthly user growth rate affects M20 revenue by 28% -- more than any other single variable. This has three implications:

1. Marketing budget allocation has higher revenue ROI than feature development
2. Product-market fit validation during Pilot 1 is the highest-value activity
3. Facility partnerships that bring patient referrals are more valuable than pricing optimization

Takeaway 3: Pharmacy and Diagnostics Add Meaningful Revenue

The v2 model explicitly includes pharmacy take-rate (8-12%) and diagnostics take-rate (10-15%) per params.md Section 11.2. These streams add 15-25% to consultation-only revenue at production scale, with virtually no additional marginal cost (the platform already routes these orders). At $350-700/mo for pharmacy and $174-348/mo for diagnostics (per 1,000 active patients), these streams provide a meaningful supplementary revenue layer that was underweighted in v1.

Takeaway 4: The Primary Path Reaches Revenue-Cost Parity at M18-M19

The S3-O2-I2 primary path reaches monthly revenue parity with monthly operating costs around Month 18-19, coinciding with the final months of the 20-month build window. This means the company transitions from a capital-burning entity to a revenue-covering entity right as the product build completes. Post-M20, revenue continues to grow while costs stabilize, closing the cumulative gap.

Takeaway 5: The Success Metric Is Achievable Within 30-32 Months for the Primary Path

The primary success metric (cumulative revenue exceeds cumulative economic cost) is projected at Month 29-32 for S3-O2-I2. This is approximately 10 months after the 20-month build window closes. During those 10 months, monthly revenue of $16,000-$30,000 (growing at 15% monthly) accumulates the remaining ~$198,000 needed to close the gap.

For the upside path (S3-O2-I3), the success metric is achievable as early as Month 23-26 due to both lower total economic cost ($251,808 vs $276,996) and faster revenue growth from full investor backing.

Takeaway 6: Avoid the Bootstrapped + Large Investor Trap

Combinations where founders self-fund at L1/O1 ($500/mo equivalent) but receive large investors ($150K+) show the worst revenue efficiency. The slow O1 development phase delays revenue start by 2-4 months, meaning investor capital sits idle during the extended Pre-Pilot. If bootstrapped self-funding is the constraint, a smaller investor (I1 or I2) is more capital-efficient than I3.

Takeaway 7: Kenya Adds 40-80% Revenue Uplift When Launched

Kenya's higher GDP per capita ($2,100 vs $1,020), higher health spending ($84/yr vs $28/yr per capita), and higher smartphone penetration (55-65% vs 35-45%) support higher ARPU and faster user acquisition. When Kenya launches (targeted for Pilot 2 or early Production per 01-timeline.md), revenue from the combined Ethiopia + Kenya operation is projected at 1.4-1.8x the Ethiopia-only figure.

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11. Cross-References

Document	Relevance to Revenue Modelling
params.md	All scenario definitions, rates, phase timing, market parameters, pricing corridors (Sections 9, 10, 11, 13, 14, 15)
01-timeline.md	Phase durations and month boundaries for all 21 paths (Section 4.3)
03-cost-breakdown.md	Monthly economic costs by phase and level (Sections 4.5, 5.2, 6) used in Section 9 overlay
04-effort-estimation.md	Development timeline driving revenue start dates
06-pricing-strategy.md	Target prices used in all revenue calculations (Sections 5.2, 6); unit economics (Section 6); competitor benchmarks (Section 8)
07-roi-analysis.md	ROI calculations built on this document's projections
08-customer-acquisition.md	User growth assumptions and marketing strategy
09-business-metrics.md	KPIs including ARPU, churn, LTV:CAC derived from revenue model
10-cash-flow-runway.md	Cash flow projections that combine this document with cost model

---

Appendix A: Revenue Model Formulas

A.1 Monthly B2C Revenue

```
B2C_Revenue = Active_Users x Consultations_Per_Active x Avg_Fee
            + Active_Users x Premium_Conversion x Subscription_Fee
            + Consultations x Video_Attach_Rate x Video_Fee
            + Consultations x Lab_Attach_Rate x Lab_Fee
            + Consultations x Rx_Attach_Rate x Rx_Fee
```

A.2 Monthly B2B Revenue

```
B2B_Revenue = Paying_Facilities x Platform_Fee
            + Facility_Consultations x Commission_Rate x Avg_Fee
            + Pharmacy_Orders x Avg_Order_Value x Pharmacy_Take_Rate
            + Diagnostics_Orders x Avg_Order_Value x Diagnostics_Take_Rate
```

A.3 Net User Growth

```
Users(t+1) = Users(t) x (1 + Growth_Rate) x (1 − Churn_Rate)
Active_Users = Registered_Users x Active_Rate
```

Net monthly growth rate = (1 + Growth_Rate) x (1 - Churn_Rate) - 1

At moderate assumptions: (1 + 0.15) x (1 - 0.07) - 1 = 1.15 x 0.93 - 1 = 0.0695 = 6.95% net monthly growth.

This means registered users roughly double every 10 months at moderate assumptions, or every 7 months at optimistic assumptions.

A.4 Success Metric

```
Success = Cumulative_Revenue(t) >= Cumulative_Economic_Cost(t)
where Cumulative_Revenue = SUM(Monthly_Revenue, M0..t)
and   Cumulative_Economic_Cost = SUM(Monthly_Economic_Cost, M0..t)
      per 03-cost-breakdown.md Section 5.2
```

A.5 ARPU Formula

```
ARPU = B2C_Revenue_Per_Active_User + (B2B_Revenue / Active_Users)
     = (Consults/Active x Avg_Fee)
     + (Pharmacy_Orders/Active x Order_Value x Take_Rate)
     + (Diagnostics_Orders/Active x Order_Value x Take_Rate)
     + (Video_Attach_Rate x Video_Fee)
     + (Lab_Attach_Rate x Lab_Fee)
     + (Rx_Attach_Rate x Rx_Fee)
     + (Premium_Conversion x Sub_Fee)
```

Appendix B: INR Conversion Reference

All revenue figures in this document are in USD. For Indian operations staff cost comparisons (per params.md Section 4), the following INR equivalents apply at 83.0 USD/INR:

Revenue Milestone (USD/mo)	INR Equivalent (per mo)	What It Covers
$500	INR 41,500	Less than 1 doctor salary (INR 60,000)
$2,482	INR 205,986	Pilot 1 ops staff (2 doctors + 2 admin + 2 tech)
$3,000	INR 249,000	Pilot 2 entry revenue target
$4,361	INR 361,963	Pilot 2 ops staff (4 doctors + 4 admin + 2 tech)
$6,542	INR 542,986	Prod Ready ops staff (6 doctors + 6 admin + 3 tech)

$10,000	INR 830,000	Approaching full ops staff + infrastructure coverage
$15,000	INR 1,245,000	Revenue covers full monthly economic cost at L2 ($14,811 per 03-cost-breakdown.md)

Appendix C: Key v1-to-v2 Revenue Model Differences

Parameter	v1 Value	v2 Value	Impact on Revenue Model
v1 baseline scenario	S2-L2-I2	S3-O2-I2 (primary)	Different investor timing; S3 provides earlier capital
Phase durations	Derived from old effort model	Derived from v2 01-timeline.md with 2,195-hour total effort	More realistic phase boundaries
Ops staff in breakeven	Not modeled	INR 206,000-543,000/mo ($2,482-$6,542)	Higher breakeven consultation volume
Pharmacy/diagnostics revenue	Mentioned but not modeled	8-12% and 10-15% take-rates explicitly modeled	+15-25% to consultation-only revenue
Monthly economic cost	$500-$2,000/mo abstraction	$10,478-$18,972/mo realistic (per 03-cost-breakdown.md)	Much higher bar for revenue to clear
Parallel ops workstreams	Not considered	750 hours consuming founder time	Explains why Pre-Pilot is 3-6 months, not 1-2
Equity structure	Not integrated	60/20/20 (per params.md Section 8)	Investor capital expectations are bounded
Android APK	Vague	480 hrs fixed at $15/hr = $7,200	Patient channel availability affects user growth timing
Web platform	682-1,413 hrs	745 hrs at $20/hr = $14,900	Tighter development timeline
Infrastructure costs	$30-50/mo flat	$65-$1,500/mo by phase	Realistic infrastructure breakeven

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End of 05-revenue-modelling.md v2. All figures trace to params.md v2.0, 01-timeline.md v2.0, 03-cost-breakdown.md v2.0, and 06-pricing-strategy.md v2.0. March 2026.